

"Future Risk Predictor": An Interactive Machine Learning Dashboard for Analyzing and Forecasting COVID-19 Excess Mortality

AN MACHINE LEARNING REPORT

*Submitted for the fulfillment
of
Machine Learning CA3: Mini Project Report*

Submitted by

Pratham Agrawal, 22070521078

B. Tech Computer Science & Engineering

pratham.agrawal.btech2022@sitnagpur.siu.edu.in

Symbiosis Institute of Technology, Nagpur Campus, Symbiosis International
(Deemed University) Pune

This Document is prepared for

Dr. Piyush Chauhan

TABLE OF CONTENTS

S. No.	Chapter	Title	Page Number
1.		Table of Contents	2
2.		Abstract	3
3.	1	Introduction	5
	1.1	Background and Context	5
	1.2	Motivation	5
	1.3	Problem Statement	5
	1.4	Objectives of the Project	5
	1.5	Novelty of Work	6
4.	2	Literature Review	7
5.	3	Methodology	10
	3.1	System Architecture	10
	3.2	Data Collection and Preprocessing	11
	3.3	Exploratory Data Analysis (EDA)	11
	3.4	Feature Engineering	13
	3.5	Model Implementation and Algorithms	14
	3.6	Tools, Libraries, and Frameworks	16
6.	4	Implementation	17
	4.1	ML Pipeline Construction	17
	4.2	Streamlit Dashboard Development	17
	4.3	Challenges and Solutions	17
	4.4	Sample Outputs (Screenshots)	18

7.	5	Results and Discussion	20
	5.1	Experimental Setup	20
	5.2	Model Performance Evaluation	20
	5.3	Interpretation of Key Findings	21
8.	6	Conclusion and Future Work	23
	6.1	Conclusion	23
	6.2	Limitations of the Current Work	23
	6.3	Future Work	23
9.		References	25
11.		Appendix	27

ABSTRACT

The COVID-19 pandemic has caused a severe health crisis worldwide. Excess mortality, defined as the number of deaths beyond what is expected, is a critical metric for measuring the severity of a situation. In this project, we illustrate how to take advantage of the WHO's comprehensive available data to ameliorate a machine learning pipeline that ultimately assesses, models and predicts excess mortality from end to end. The main modelling technique followed a multi-task approach, first taking on a Regression task to provide a prediction of the total excess deaths; producing a Classification task which assigned demographic groups a "Low", "Medium" or "High" level of risk; and finally, undertaking a Clustering task which produced latent profiles of country characteristics. There were many important findings in our evaluation, but the most accurate models were the Gradient Boosting Regressor given R-squared of 0.87 and MLP Classifier given an accuracy of 0.91. The overall output of the project is the "Future Risk Predictor", a Streamlit interactive dashboard that brings together all of the models for practically analysing and assisting the development of informed decisions through the complexities of predictive outcomes.

Keywords: Machine Learning, Excess Mortality, COVID-19, Gradient Boosting, MLP Classifier, Streamlit, Data Visualization

CHAPTER 1

INTRODUCTION

1.1 Background and Context

The COVID-19 pandemic, caused by the SARS-CoV-2 virus, is one of the most significant global public health emergencies in recent memory. One way to view its impact is through officially confirmed deaths but "excess mortality" is a better measure. The World Health Organization (WHO) dataset used for this project also has this measure that captures the total number of deaths that have occurred beyond what would be expected in normal times, including those deaths directly or indirectly due to the pandemic (for example, due to overwhelmed healthcare systems).

1.2 Motivation

The complex global dataset from WHO and raw data is large. The intention of the project wasn't simply to conduct a basic historical analysis, but to develop this raw data into something useful. It is important for vulnerable groups not only to understand what has happened but also understand to predict what may happen in similar situations.

1.3 Problem Statement

The main difficulty lies in analyzing the complex, high-dimensional WHO Excess Deaths dataset to adequately surface and singularize causes of death and to construct accurate prediction models. The first of two key aspects of this difficulty is identifying the best machine learning models to predict death counts (as a regression problem) and determine risk (as a classification problem). The second aspect is to make contextually-sound predictions understandable to a non-specialist.

1.4 Objectives of the Project

To resolve the stated problem, the project was informed by guiding objectives. First, EDA was performed on the WHO dataset to reveal important international, regional and demographic trends in excess mortality. Following the analysis, the next objective was to train and validate a collection of machine learning models for three distinct tasks: i) regression model that predicts the numerical value of excess deaths, ii) classification model that predicts the categorical risk level (Low, Medium, High), and iii) clustering model that reveals non-obvious groupings or "profiles" of

countries. The next step was to select the most effective models for each predictive task, which required some critical comparisons among models using the appropriate metrics, including R-squared, RMSE, and Accuracy. Ultimately, the final goal was to create and present a "Future Risk Predictor" as an interactive Streamlit web application, which leverages the most effective models combined into a usable, easy-to-use tool for users.

1.5 Novelty of Work

While machine learning has been exploited across the literature as a method for examining COVID-19 data, often related to clinical diagnosis or case prediction, this work is unique in that it utilizes a multi-model approach across the entire study. The multi-model approach combines regression and classification models, and provides a seamless, interactive dashboard approach to monitor population-based excess mortality. As such, a dashboard is produced that connects raw data to useful predictive insights.

CHAPTER 2

LITERATURE REVIEW

A survey of the literature reveals that machine learning has been widely applied to pandemic data. However, the majority of research focuses on epidemiological data for case-count forecasting or clinical data for patient-level prognosis. This project applies approaches from related fields to the specific problem of excess mortality at the macro level. The empirical review of existing methods is presented in Table 1.

Table 1. Empirical review of existing methods

Reference	Method Used	Findings	Results	Limitations
[1] Msemburi, W., et al. (2023)	Bayesian Poisson framework to model ACM and expected deaths.	Estimated global excess deaths for 2020-21 and provided the official WHO methodology.	Estimated 14.83 million global excess deaths.	Paucity of data in some regions (e.g., Africa).
[2] Karlinsky, A., & Kobak, D. (2021)	Regression model on 2015-2019 data to project expected deaths.	Created the World Mortality Dataset, a key resource for tracking excess mortality.	Assembled a comprehensive, publicly available dataset for analysis.	Relies on the availability of national vital statistics.
[3] Ferenci, T. (2023)	Comparison of spline models, linear trends, and averages.	Spline models are suitable for baseline prediction but are highly sensitive to parameterization, creating a risk of overfitting.	Performance of spline models is highly dependent on parameter choice.	Based on synthetic data scenarios, not real-world comprehensive data.
[4] Liu, J., et al. (2021)	Compared XGBoost, Logistic Regression (LR), SVM, and Random Forest (RF) for Acute Kidney Injury (AKI) mortality prediction.	XGBoost demonstrated superior performance in a clinical setting for predicting AKI mortality.	XGBoost achieved the highest AUROC (0.796) and accuracy (0.860).	Single-center study using the eICU database, which may limit generalizability.

[5] Khatami, S., et al. (2024)	Meta-analysis of Machine Learning (ML) models for post-Myocardial Infarction (MI) mortality prediction.	Advanced Gradient Boosting Machines (GBMs), particularly XGBoost, were identified as the most reliable models.	Synthesized evidence showed XGBoost achieved an AUC of 0.90.	High heterogeneity was observed across the reviewed studies.
[6] HyeYoung Lee et al. (2025)	Random Forest with SHAP (SHapley Additive exPlanations) for all-cause in-hospital mortality.	Demonstrated the use of SHAP for enhancing model interpretability in a clinical context.	Identified Age, lymphocytes, and neutrophils as the key predictors of mortality.	Single-database study (MIMIC-III), which may affect the generalizability of the findings.
[7] Wang, Z., et al. (2024)	Compared DeepSurv, CoxTime, DeepHit, RSF, GBM, and Cox models for survival analysis.	Deep learning models, specifically DeepSurv, achieved the best discrimination and calibration among the tested models.	DeepSurv achieved the best C-index (0.893) and Integrated Brier Score (IBS) (0.041).	The analysis was retrospective and conducted at a single center.
[8] Pourhomayoun, M., & Shakibi, M. (2021)	Comparative study of various ML models (DT, RF, SVM, etc.) for COVID-19 mortality.	Provided a broad comparison of multiple algorithms for predicting mortality risk in an acute public health crisis.	The study served as a comparative benchmark for various ML models on early pandemic data.	The findings are specific to early pandemic data and may not apply to later stages.
[9] Nigam, A., et al. (2024)	Comparative analysis of ANN, DT, KNN, and Naive Bayes for Road Traffic Accident (RTA) mortality.	Showcased the power of ensemble methods, which outperformed individual base classifiers.	Ensemble methods achieved 89% accuracy.	The study is specific to mortality from Road Traffic Accidents (RTAs).

[10] Torres, R. S., et al. (2024)	Support Vector Machine (SVM) to classify causes of death.	Proposed a novel framework using SVM to address data quality issues in mortality data.	The model can be used to classify data for countries where causes of death are misreported.	The focus is on the classification of causes of death, not the prediction of mortality rates.
[11] Loe, S. S., et al. (2024)	Random Forest for Recurrent Events based on Pseudo-Observations (RFRE.PO).	Developed a powerful and novel tool for the dynamic risk prediction of recurring health events.	Advanced the application of Random Forest for dynamic and personalized predictions.	The model was tested on a specific disease (COPD), and its applicability to other conditions is not yet established.

The use of models such as MLPs (ANNs) and Gradient Boosting (XGBoost) is validated in the literature as practical tools. Nevertheless, there is a big gap in combining these various predictive tasks (i.e., classification for risk level and regression for quantity) into a single, easily usable tool for researchers or policymakers, particularly when using the extensive WHO global excess deaths dataset. By developing an end-to-end system that transitions from data analysis to a functional, predictive Streamlit dashboard, our project fills this gap.

CHAPTER 3

METHODOLOGY

This project is designed as an end-to-end pipeline that converts unprocessed data into an interactive, predictive tool. A three-task machine learning strategy was the primary method. Regression was used in the first task to forecast the excess mean, or continuous numerical value of excess deaths. Predicting a categorical risk level of "Low," "Medium," or "High" was the goal of the second task, classification. The third task, Clustering, used unsupervised learning to uncover obscure, non-visible country profiles. The final "Future Risk Predictor" dashboard was created by integrating these models into a Streamlit web application.

3.1 System Architecture

Figure 1 illustrates the project workflow in a sequential, multi-stage architecture. Initial data acquisition is followed by cleaning and feature engineering in the pre-processing stage. Later, the prepared data is used to train the three parallel machine learning tasks, as well as for exploratory analysis. After training, the performance of the supervised models will be evaluated to select the "winning" model in both regression and classification. Optimized models are integrated into the backend feed, which powers the user-facing dashboard of the Streamlit application.

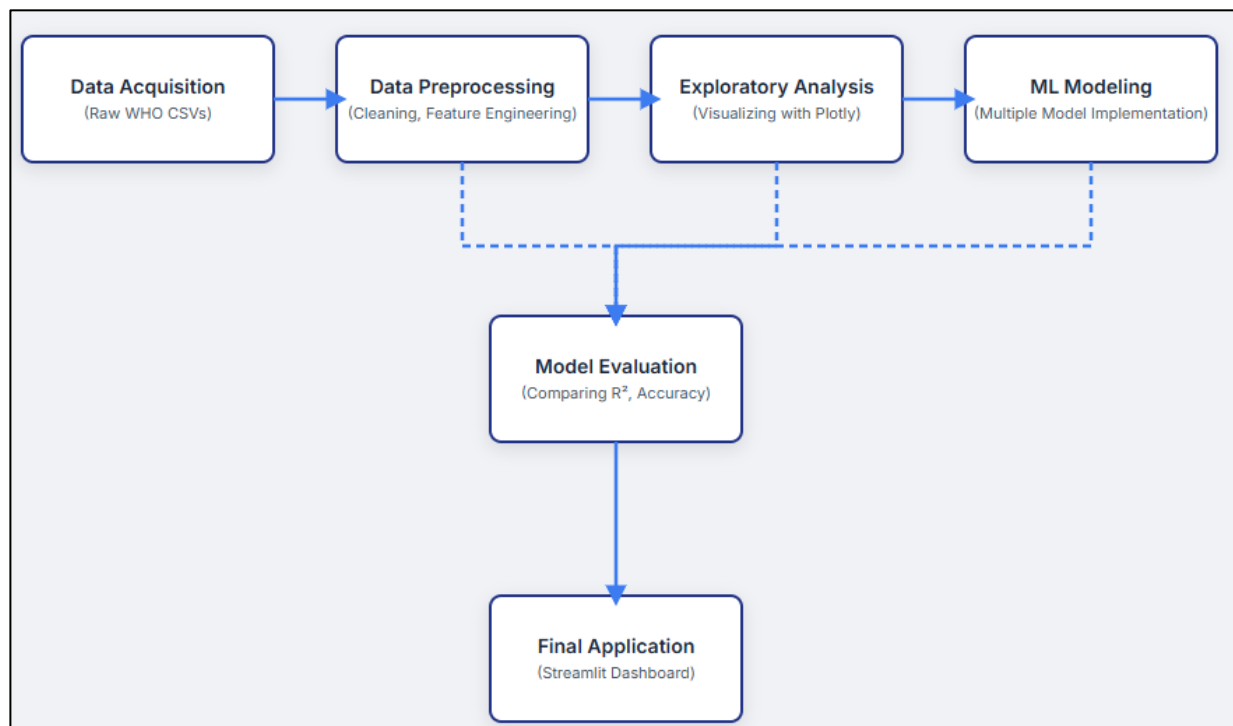


Figure 1: Our Project Pipeline

3.2 Data Collection and Preprocessing

For the purposes of further manipulation of the data, the raw dataset file was first opened in a Pandas DataFrame. This was then followed by the data cleaning process. In this phase, only the relevant features to be analyzed subsequently were included, these being: country, ISO 3, year, sex, age group, excess mean, ACM mean, expected mean. Any rows which had null or missing values were removed to ensure stability of the model and sound analysis. From the cleaned and analysis-friendly DataFrame, the cleaned dataset files were created, that would then be used for all machine learning models and for the development of the Streamlit application. Figure 2 shows raw dataset before preprocessing.

```
print(df_raw.head())
```

[3.1] First 5 Rows of the Raw Dataset:

	country	iso3	year	sex	age_group	type	expected.mean
0	Afghanistan	AFG	2020.0	Female	0-24	predicted	49083.643934
1	Afghanistan	AFG	2020.0	Female	25-34	predicted	6452.967039
2	Afghanistan	AFG	2020.0	Female	35-44	predicted	6117.873106
3	Afghanistan	AFG	2020.0	Female	45-54	predicted	7711.689531
4	Afghanistan	AFG	2020.0	Female	55-64	predicted	10061.544157

	acm.mean	excess.mean*
0	49103.143153	0.000000
1	6691.247219	236.607817
2	6977.363939	860.300714
3	9330.217317	1621.571806
4	12457.985086	2401.488971

Figure 2: Raw Dataset before preprocessing

3.3 Exploratory Data Analysis (EDA)

After the initial data cleaning, comprehensive exploratory analysis was carried out in order to identify underlying trends in the data. This analysis was used to inform the later stages of feature engineering and model selection. Geographical visualizations that were initially identified as the first significant discovery were that the effects of the pandemic were not spread equally worldwide. Rather, it was noted that high excess mortality was concentrated in particular countries such as India, the Russian Federation, the United States and Brazil. The second important lesson was associated with the demographics, namely the age. It was analyzed that there was a sharp correlation between age and mortality with the risk being high in the older age groups. Most of the surplus mortality was found in the age group of 65 years and above. Lastly, the sex-specific comparison of mortality revealed that there was indeed an apparent and significant difference, with the number of excess deaths being significantly higher among males than females across most areas and age

groups. The three main conclusions of the predictive significance of country, age, and sex were the main ones in justifying what features needed to be kept and encoded to the machine learning models. Figure 4, Figure 5 and Figure 6 highlight some of the important data patterns found while performing EDA Analysis on the dataset.

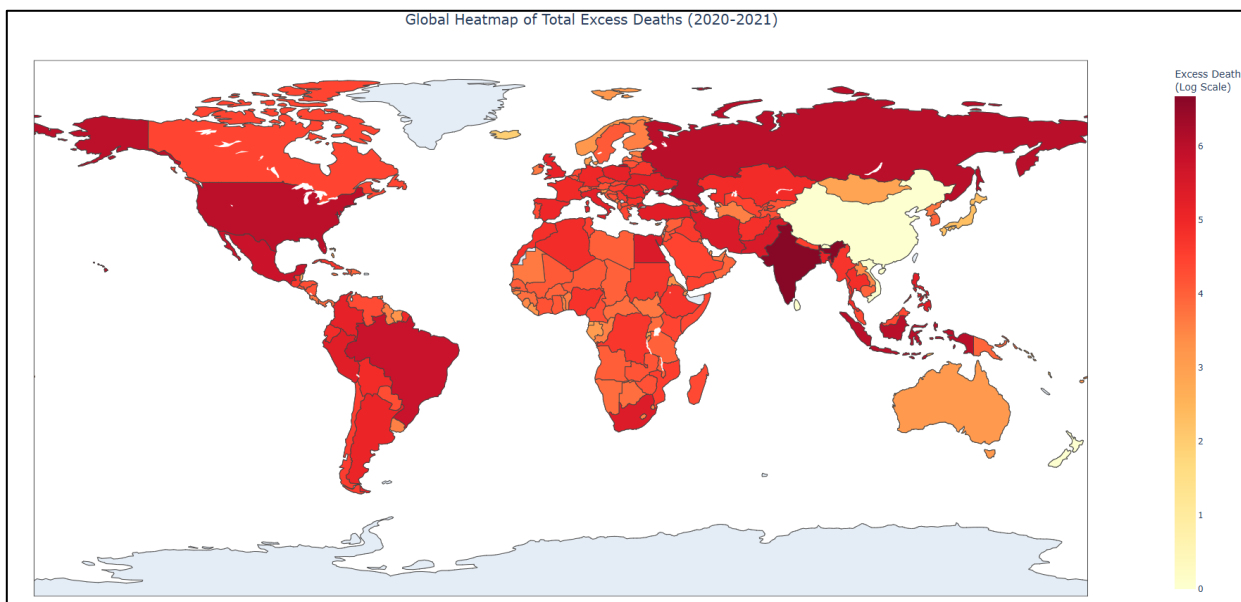


Figure 4: Global Distribution of Excess Deaths.

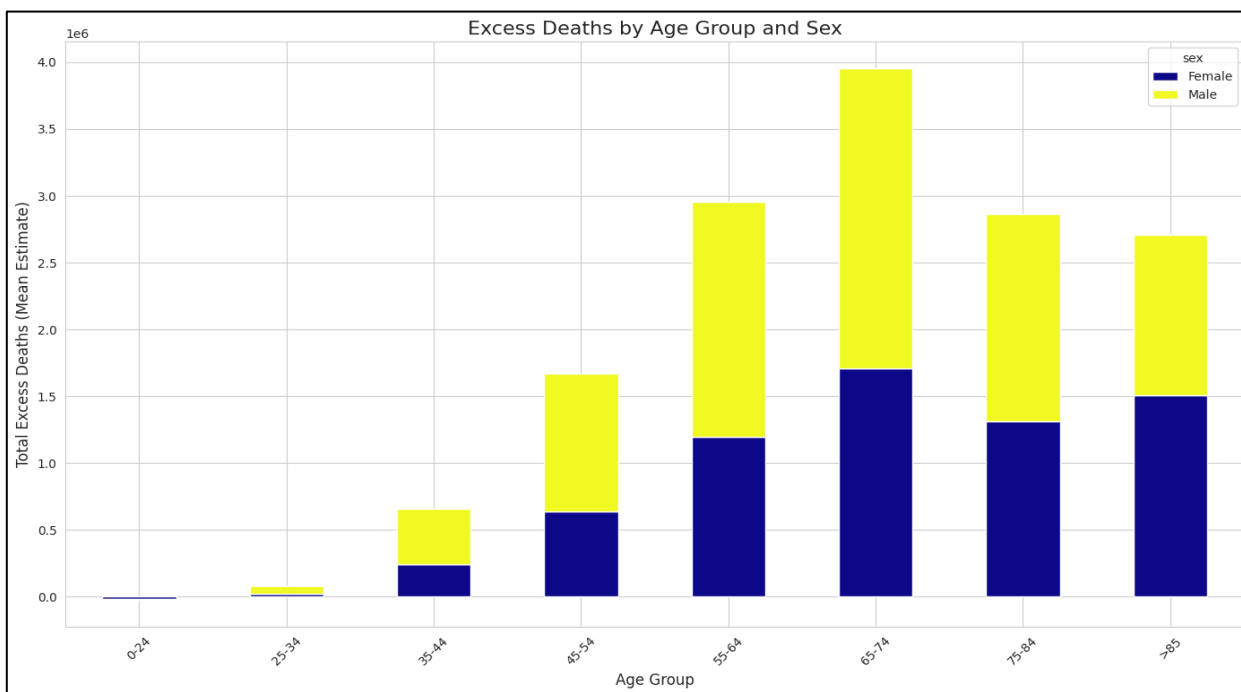


Figure 5: Global Distribution of Excess Deaths by Age Group and Sex.

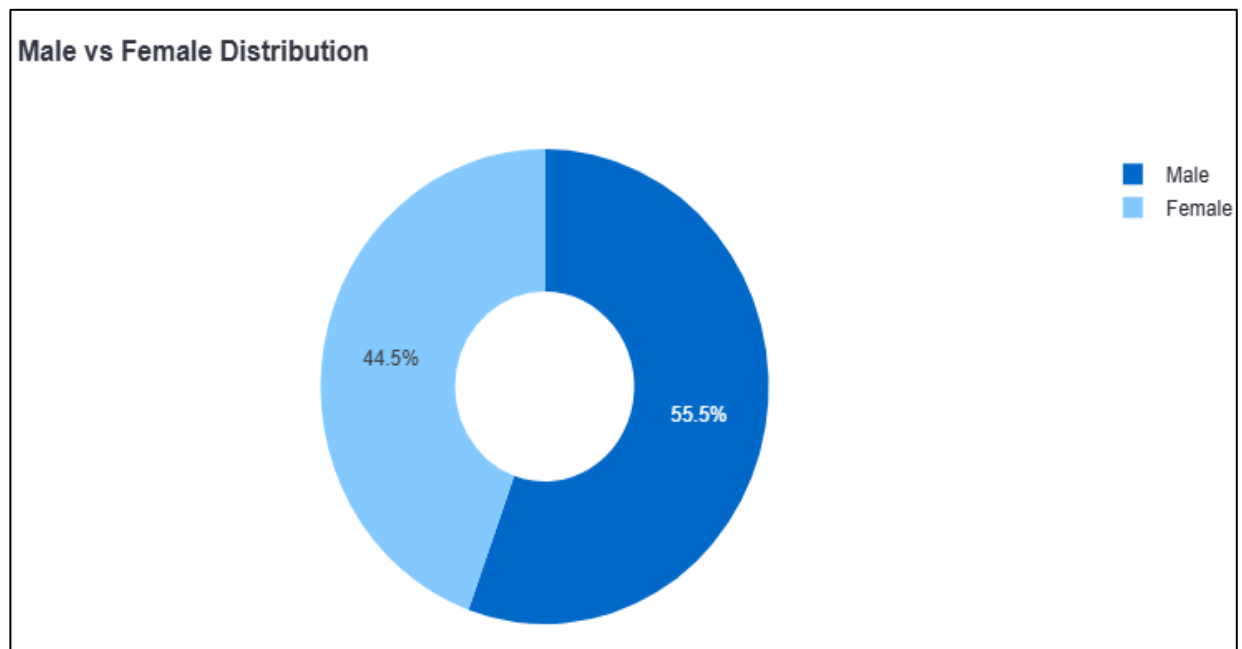


Figure 6: Male vs Female Excess Deaths Distribution.

3.4 Feature Engineering

A number of key feature engineering processes were applied in order to prepare the data for machine learning purposes. The input features, or predictors, were country, ISO3, year, sex, and age_group. Text-based, categorical features are not interpretable for machine learning models, and are encoded using a One-Hot Encoder. Given most machine learning methods cannot assume an incorrect ordinal relationship between categories, a One-Hot Encoder creates new binary columns for each unique value contained within the dataset.

Dataset 'cleaned_who_excess_deaths.csv' loaded successfully.

'risk_level' column created for later use.

	country	iso3	year	sex	age_group	type	expectedmean	acmmean	excessmean	risk_level_q	risk_level
0	Afghanistan	AFG	2020	Female	0-24	predicted	49083.643934	49103.143153	0.000000	0.333320	Medium
1	Afghanistan	AFG	2020	Female	25-34	predicted	6452.967039	6691.247219	236.607817	0.513873	High
2	Afghanistan	AFG	2020	Female	35-44	predicted	6117.873106	6977.363939	860.300714	0.666908	High
3	Afghanistan	AFG	2020	Female	45-54	predicted	7711.689531	9330.217317	1621.571806	0.667340	High
4	Afghanistan	AFG	2020	Female	55-64	predicted	10061.544157	12457.985086	2401.488971	0.667781	High

Figure 7: Data Structure after cleaning and feature engineering

As previously mentioned, both tasks in this study required a different engineering of the target variable. The excess mean column initially used in a continuous numerical format served as the target variable (y_{reg}) for the regression task. From the excess mean column, a new target feature, risk_level, was created for the classification task. The excess mean values were binned into three equal-sized groups using the Pandas qcut function, and these were subsequently labelled "Low,"

"Medium," and "High." This feature creation transformed the regression problem into a balanced classification problem. Then, a LabelEncoder was used to encode the risk_level target to then train the model. Figure 7 shows dataset after cleaning, preprocessing and feature engineering.

3.5 Model Implementation and Algorithms

For each task, multiple models were trained and evaluated to conduct a comparative analysis.

3.5.1 Regression (Predicting excessmean)

Out of the four models used, notably Linear Regression, Decision Tree Regressor, XGBoost Regressor, and Gradient Boosting Regressor, the Gradient Boosting Regressor (GBT) was the best performing model. GBT is a very strong ensemble method which creates models sequentially and, in a step,-wise fashion. It means we specify a large number of base ("weak") decision trees, each trained to correct the mistakes of previous trees. This method of correcting mistakes allows the model to identify very complicated, non-linear forms in the data. As identified in the Results section above, it was chosen based on the accuracy and credibility of predicting the requested number of deaths, as reflected by the highest R-squared value and lowest Root Mean Squared Error. Figure 8 shows regression line for actual vs predicted values using the Gradient Boosting algorithm.

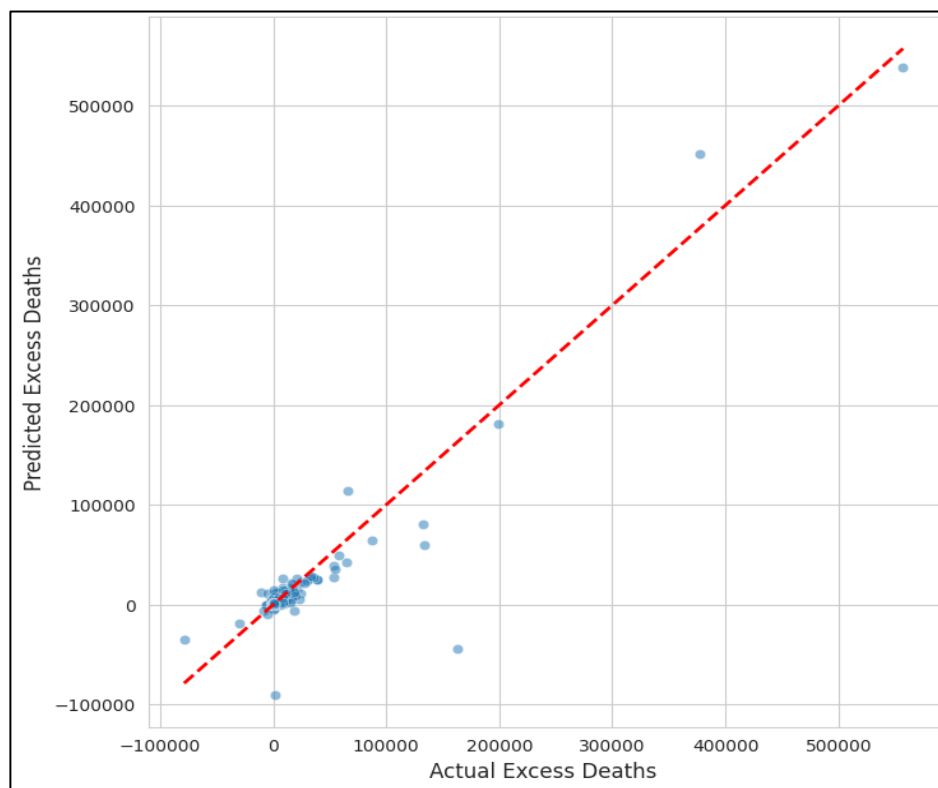


Figure 8: Regression line of Gradient Boosting Regression on actual vs predicted values

3.5.2 Classification (Predicting risk_level)

We examined four models for the classification task: Logistic Regression, K-Nearest Neighbor Classifier, Support Vector Classifier (SVC), and MLP Classifier. The MLP Classifier was the best performing. An MLP is a simple type of feedforward artificial neural network. It consists of an input layer, output layer, and multiple hidden layers. Once it ingested the one-hot encoded features, it predicted the classes 'Low', 'Medium', or 'High' using the output layer. Here the hidden layers are important, because they are able to learn complex non-linear relationships and interactions that simpler models would not be able to learn. We selected the MLP Classifier based on both the error rate related to accurately classifying the risk level of a group of individuals; it performed exceedingly well. Figure 9 shows ROC curve of our best model that is MLP vs our baseline model that is Logistic Regression.

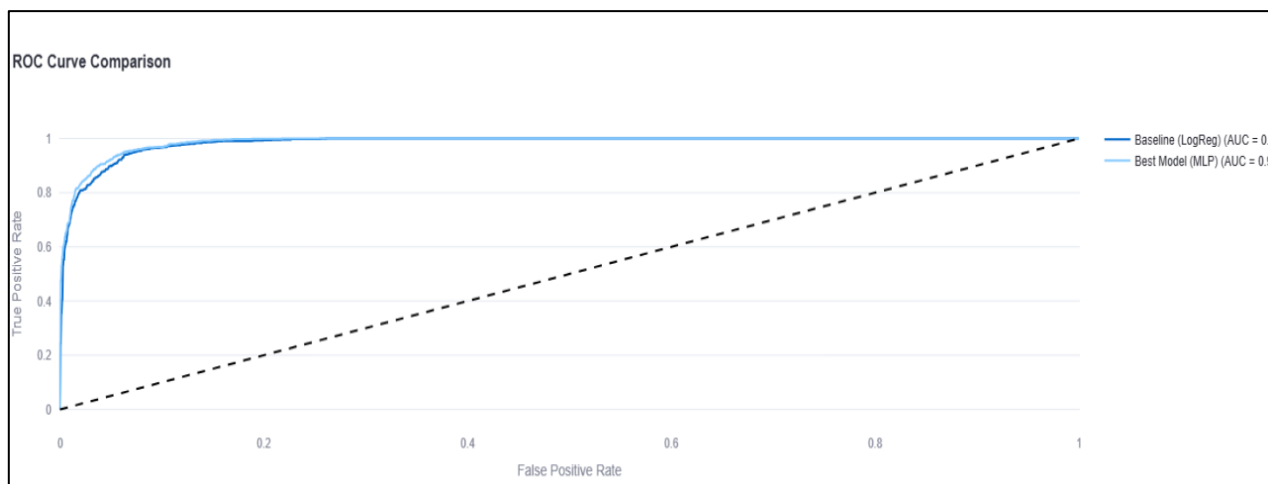


Figure 9: ROC Curve of Best (MLP) vs. Baseline (LogReg)

3.5.3 Unsupervised Clustering

The K-Means algorithm was applied to the unsupervised learning task. The goal of the K-Means algorithm is to divide a dataset into k unique, non-overlapping clusters. Each data point is iteratively assigned to the closest "centroid," or cluster center, and the centroid's location is then recalculated using the mean of the points assigned to it. K-Means was used in this project to identify "country profiles" based on two characteristics: the percentage of deaths in the senior population (65 years and older) and the total excess mortality. This helped us to identify discrete groups of countries with same kind of mortality trends. Figure 10 shows country clustering based on common mortality patterns shown between them.

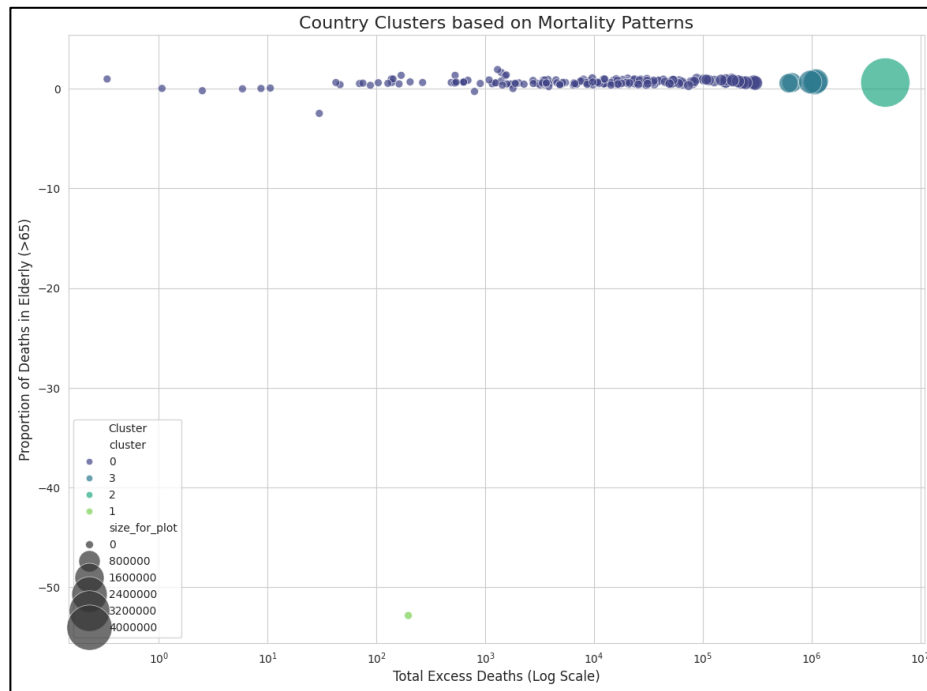


Figure 10: Shows Country Clustering based on Mortality Patterns.

3.6 Tools, Libraries, and Frameworks

This project utilized several open-source libraries and was developed entirely in Python. For data analysis and manipulation, the Pandas and NumPy libraries were used. Scikit-learn was the main machine learning library for building machine learning models and all metrics related to their evaluation. Visualizations were created using Matplotlib for static figures and Plotly for interactive charts. Lastly, the web app was designed and hosted with Streamlit.

CHAPTER 4

IMPLEMENTATION

4.1 ML Pipeline Construction

A critical aspect of this implementation is the use of scikit learn pipeline, which makes the modelling process far more efficient. The one-hot encoder was used selectively on the categorical data columns with the column transformer. The preprocessor is combined with the regressor model to form a single Pipeline object. This is significant because it guarantees that the preprocessing will be executed the same way, either on the training data or on new data, from the user, from the dashboard. This step prevents data leakage and ensures predictions that are both reliable and consistent.

4.2 Streamlit Dashboard Development

The Streamlit framework was utilized to build the final, optimized models into an actual web app. The app file is stored inside the app.py file, and it is modular having multiple pages that can be accessed through a sidebar. The About page describes the project, the Global EDA page visualizes global trends, the Comparative Analysis page has a more detailed overview, and the ML Analysis page displays the results for the K-Means clustering and classifier evaluation.

The last component of the project is the "Future Risk Predictor" page. The Future Risk Predictor page has selection boxes for Country, Year, Sex, and Age Group. Upon clicking the "Predict" button, the user will feed the user chosen filters to the program and the program will generate a data frame. The Pipeline objects of the MLP Classifier and GBT Regressor are loaded with the data frame and updated with the saved ones. The pipelines will produce forecasts and will automatically preprocess the input. The findings are displayed on the dashboard as a MLP Model Predicted risk level and GBT Model Predicted Excess deaths.

4.3 Challenges and Solutions

There were a number of challenges that were encountered in the development. First, sex, age, and geography relations are complex and non-linear and could not be well represented by simple models including the linear and logistic regression. The solution was to test deep learning models, such as the MLP, and more complex ensemble models, such as Gradient Boosting, in depth. The results

proved that these advanced models had a lot of positive effect on performance. The second concern was that an individual would not comprehend one numerical forecast, such as 110,000 deaths because it would not be in context. To handle this we created a feature of risk level and modelled a classification to do so. This allows the dashboard to deliver two kinds of information, a qualitative assessment based on classification, which is more easily interpretable, and a quantitative assessment of regression.

4.4 Sample Outputs (Screenshots)

The following figures (Figure 11, Figure 12, Figure 13, Figure 14, Figure 15) are screenshots that showcase the main features of our finished Streamlit application.



Figure 11: Shows Global EDA page of our streamlit dashboard

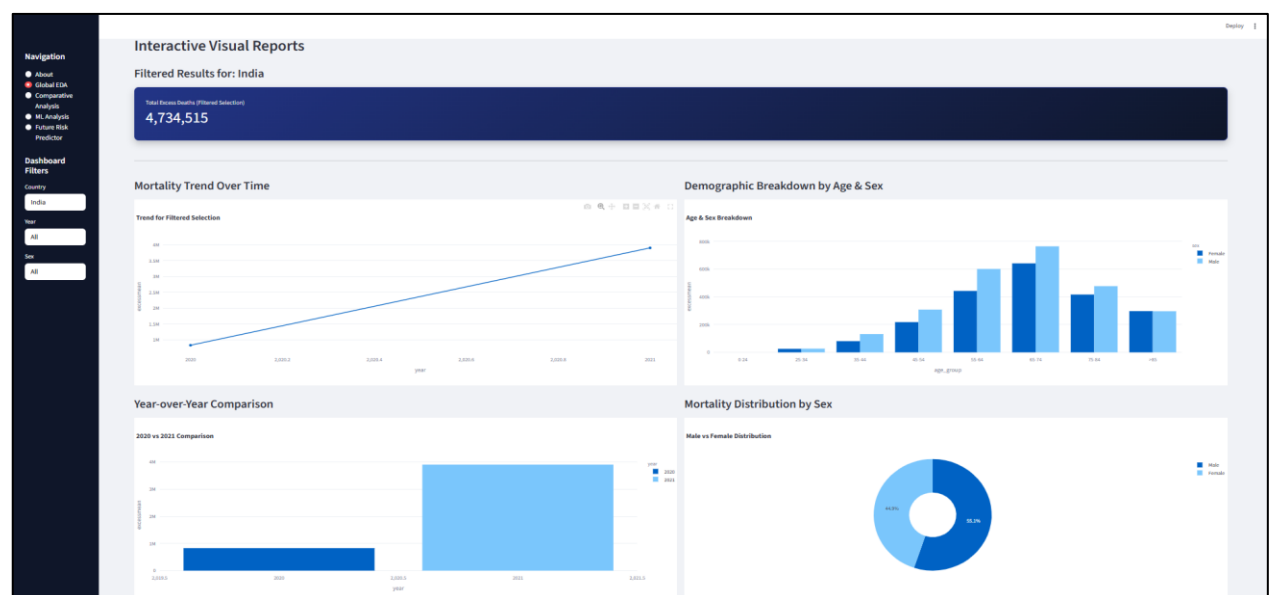


Figure 12: Shows Interactive Visual Reports feature of our dashboard

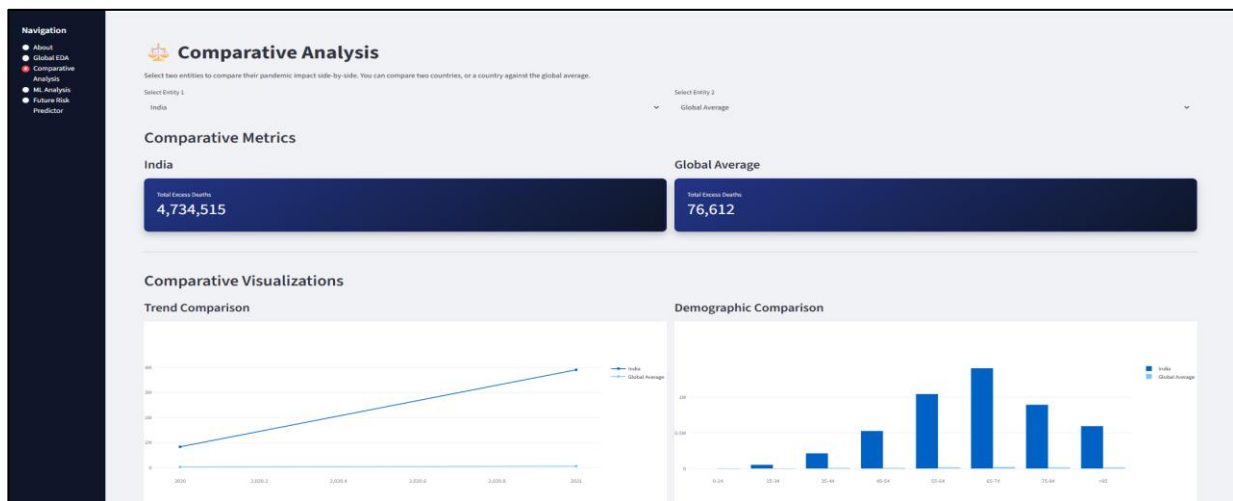


Figure 13: Shows page where comparative analysis of different countries based on filters

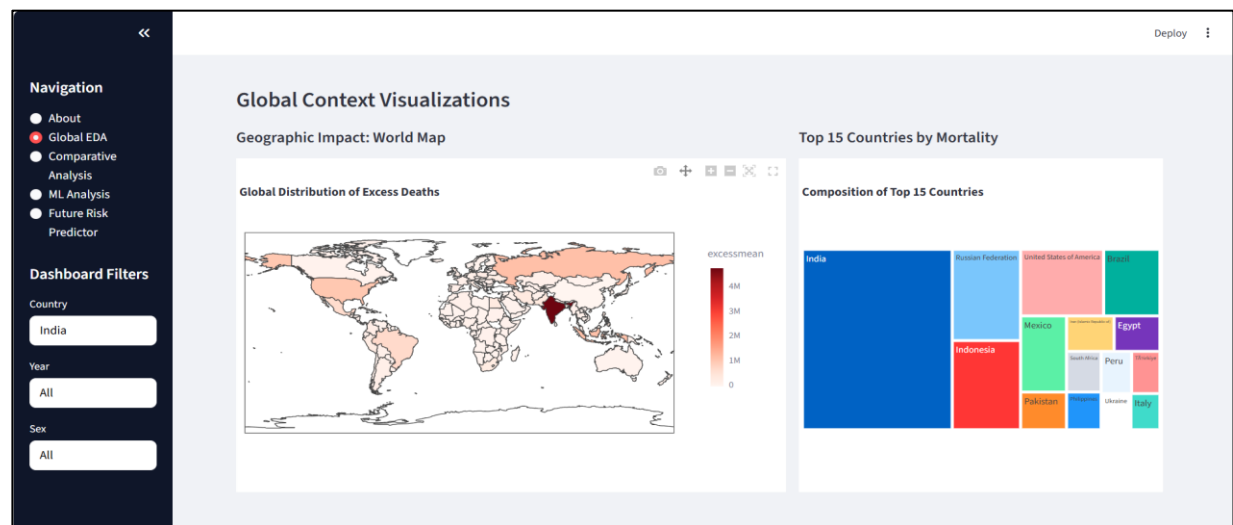


Figure 14: Shows additional visualizations related to global context

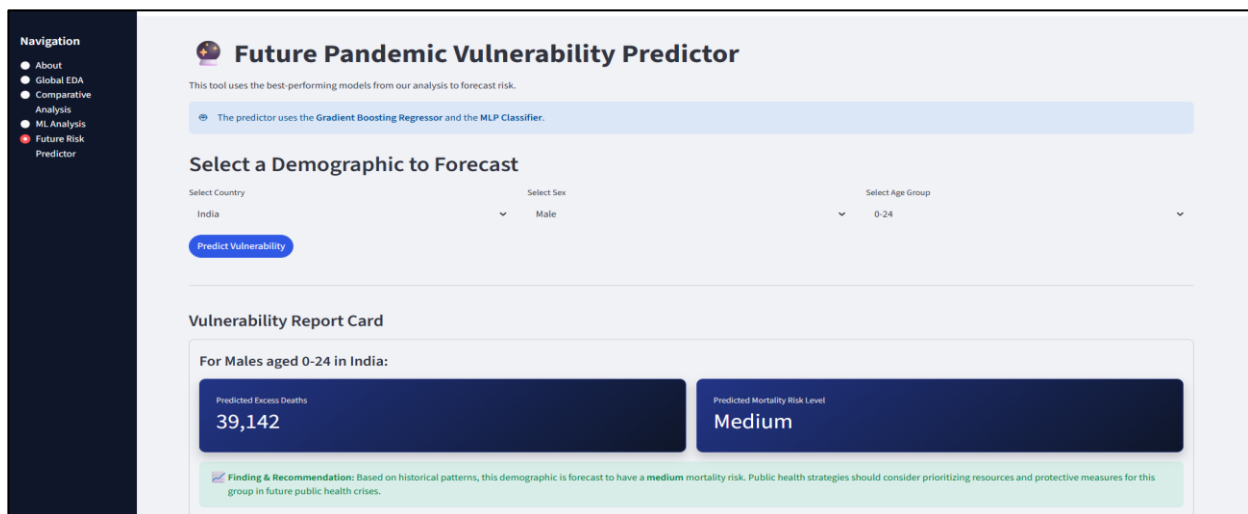


Figure 15: Main feature of our dashboard the future vulnerability predictor

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Experimental Setup

The project was developed using Python 3 environment. All data processing and model training were done on a typical consumer-grade computer. The primary libraries that were used were Streamlit, Pandas, scikit-learn, plotly, and were. To get the reliable model assessment when performing the task of machine learning, the dataset was separated into training and testing groups in the 80/20 proportion.

5.2 Model Performance Evaluation

The results are founded on a quantitative comparison of the machine learning models.

Table 3. Evaluation metrics of all regression models applied

Sr No.	Model	R-squared (R ²)	Mean Absolute Error (MAE)	Mean Squared Error (MSE)	Root Mean Squared Error (RMSE)
0	Gradient Boosting Regressor	0.871831	1993.1554	6.20E+07	7875.273
1	XGBoost Regressor	0.869614	1533.6747	6.31E+07	7943.0986
2	Decision Tree Regressor	0.859785	1407.9818	6.78E+07	8237.0433
3	Random Forest Regressor	0.8199	1426.4392	8.71E+07	9335.3524
4	Multiple Linear Regression	0.461381	3318.1011	2.61E+08	16144.158
5	Lasso Regression	0.461193	3299.8713	2.61E+08	16146.962
6	Ridge Regression	0.45124	3314.1516	2.66E+08	16295.415
7	MLP Regressor (Neural Network)	0.349018	2187.9241	3.15E+08	17748.383
8	Simple Linear Regression	0.005451	4078.9761	4.81E+08	21937.52
9	K-Neighbors Regressor	-0.002659	3841.2617	4.85E+08	22026.783

As far as the regression is concerned, the performance measures that were employed were R2 Score, MAE, MSE, and RMSE. These results are recapped in Table 2 and they clearly show the effectiveness of the ensemble methods. Therefore, the Gradient Boosting Regressor had the highest R2 of 0.8718 which means that it was capable of explaining more than 87% of the data. Comparatively, the Decision Tree was indicated to have high variance and the baseline linear regression model was unable to cope effectively when modeling.

Table 3. Evaluation metrics of all classification models applied

Sr no.	Model	Accuracy	Precision (Weighted)	Recall (Weighted)	F1-score (Weighted)
0	MLP Classifier (Neural Network)	0.909823	0.909289	0.909823	0.909335
1	Support Vector Classifier (SVC)	0.877617	0.877677	0.877617	0.877214
2	Logistic Regression	0.876812	0.876086	0.876812	0.876234
3	Random Forest Classifier	0.747987	0.756164	0.747987	0.749279
4	K-Neighbors Classifier	0.63124	0.64154	0.63124	0.633098

Performance on the classification task was gauged using accuracy, precision, recall and F1-score. Table 3 displays the findings. The MLP Classifier (Neural Network) with an accuracy of 0.9098 has come out as the decisive winner although the other more traditional models like SVC and Logistic Regression also performed well. This depicts its advanced ability to identify the complex trends that differentiate the Low, Medium and High risk groups.

5.3 Interpretation of Key Findings

The result of the exploratory data analysis created the basic context of the project. The illustration of excess deaths around the globe in the figure 2 illustrated that the mortality was not distributed evenly but rather highly concentrated in the specific regions, with the United States, Russia, and India having the most cumulative increase. The most important risk factor, as shown by further analysis of demographic data, was age with the vast majority of the excess deaths being found in

populations above 65. Another significant difference was in the gender gap where men were more likely to experience excess mortality in most countries and age groups. These early findings were greatly extended with the machine learning models. As an example, we were capable of seeing beyond pure totals, owing to our K-Means clustering. It was able to automatically classify the countries based on their mortality patterns as in Figure 3. We found a group of countries with a large number of deaths but we also found an interesting group with a low number of deaths yet a high number of deaths among the elderly. This helps us to draw comparisons in the impact on different countries in a better informed way. At the same time, we tested our best classification model, which is the MLP Classifier. We were convinced that the model was not just conjectural by the fact that its ROC curve showed that the AUC of the model was high at all the three risk levels.

We eventually took all this and made it into our Future Risk Predictor as it is represented in Figure 5. It is this dashboard which is the actual end product of all our work. It is a hybrid of our two favorable models, the MLP Classifier and the Gradient Booster, built into a simple and straightforward utility. When you put in a demographic, you receive two answers: a text description of the estimated level of risk and a hard number of the number of expected deaths. The main aim of the project was to compress all of this complex data and all these trained models into one single understandable conclusion that any person could utilize.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The project was a successful design, implementation and evaluation of an end-to-end machine learning pipeline to analyse and predict excess COVID-19 mortality using WHO dataset. The exploratory analysis revealed significant trends which validated the point that risk was not evenly distributed, but was concentrated by sex, age and place. The Gradient Boosting Regressor turned out to be the most appropriate model of machine learning to estimate the number of fatalities after a comprehensive comparison with the models provided a registered R-squared of 0.87. The most appropriate model to calculate the level of risk was the MLP Classifier, which received the accuracy level of 0.91. An interactive Streamlit dashboard, the Future Risk Predictor, is a new product of this work, an integration of the two successful models. This tool is also able to solve the problem of providing the non-technical audience with a better insight into the complex world of predictive modelling, by simply filling the gap between unprocessed data and valuable insights.

6.2 Limitations of the Current Work

This study has a number of limitations. As the models were trained on the data of 2020 and 2021, they may not be able to explain the effect of new viral strains, mass vaccination, or changes in the rules governing the work of the population. Also, there are only basic demographics and other potentially important predictors like population density, expenditure on healthcare, and comorbidity rates do not appear in the models. Finally, but importantly, the classification of risk level is statistical quantiles, not extrinsic epidemiological definitions, although, as mentioned, useful.

6.3 Future Work

The future development has a few glaring avenues. The most obvious and the first improvement would be to increase the amount of data that the models have. Presently, they are relying on simple demographics. It is likely that we would get even more precise predictions when we added economic data, e.g., GDP per capita, the data regarding the healthcare system in each country, or the overall health statistics, e.g. the prevalence of obesity or diabetes. However, the other interesting alternative would be to modify the kind of prediction. Rather than merely forecasting the fixed number of mortality in a given year, it would be extremely helpful to employ time-series models,

including ARIMA or LSTM, to predict the trends in mortality in the next several months. To develop a live tool, which can be used by any researcher or policymaker as a cloud service.

REFERENCES

- [1] W. Msemburi et al., "The WHO estimates of excess mortality associated with the COVID-19 pandemic," *Nature*, vol. 613, no. 7942, pp. 130–137, Jan. 2023.
<https://doi.org/10.1038/s41586-022-05522-2>
- [2] A. Karlinsky and D. Kobak, "Tracking excess mortality across countries during the COVID-19 pandemic with the World Mortality Dataset," *eLife*, vol. 10, p. e69336, 2021.
<https://doi.org/10.7554/eLife.69336>
- [3] T. Ferenci, "Comparing methods to predict baseline mortality for excess mortality calculations," *BMC Medical Research Methodology*, vol. 23, no. 1, p. 239, Oct. 2023.
<https://doi.org/10.1186/s12874-023-02061-w>
- [4] J. Liu, J. Wu, S. Liu, M. Li, K. Hu, and K. Li, "Predicting mortality of patients with acute kidney injury in the ICU using XGBoost model," *PLoS ONE*, vol. 16, no. 2, p. e0246306, Feb. 2021.
<https://doi.org/10.1371/journal.pone.0301014>
- [5] S. Khatami et al., "Comparing Machine Learning Models for Predicting Mortality after Myocardial Infarction: A Systematic Review and Meta-Analysis," *Archives of Academic Emergency Medicine*, vol. 12, no. 1, p. e1, 2024.
<https://doi.org/10.22037/aaem.v14i1.2783>
- [6] HyeYoung Lee, Pavel Tsoi, "Feature-Enhanced Machine Learning for All-Cause Mortality Prediction in Healthcare Data," *arXiv preprint arXiv:2503.21241*, Mar. 2025.
<https://doi.org/10.48550/arXiv.2503.21241>
- [7] Z. Wang et al., "Survival modeling using deep learning, machine learning and statistical methods: A comparative analysis for predicting mortality after hospital admission," *arXiv preprint arXiv:2403.06999*, Mar. 2024.
<https://doi.org/10.48550/arXiv.2403.06999>
- [8] M. Pourhomayoun and M. Shakibi, "Predicting mortality risk in patients with COVID-19 using machine learning," *Informatics in Medicine Unlocked*, vol. 22, p. 100478, 2021.
<https://doi.org/10.1016/j.smhl.2020.100178>
- [9] A. Banerjee, J. J. Jena, M. Kumar Mishra, A. R. Panda and S. Darshana, "Comparative Analysis of Machine Learning and ANN models for Mortality prediction in RTAs," 2023

OITS International Conference on Information Technology (OCIT), Raipur, India, 2023, pp. 698-702, doi: 10.1109/OCIT59427.2023.10431379.

<https://doi.org/10.1109/OCIT59427.2023.10431379>

- [10] R. S. Torres, C. Camarda, and V. Kaishev, "An Alternative Approach to Causes of Death Prediction Using Support Vector Machines," ResearchGate, 2024. [Online]. Available: <https://www.researchgate.net/publication/394117043>.
- [11] S. S. Loe, S. Murray, and M. Wu, "Random forest for dynamic risk prediction of recurrent events: a pseudo-observation approach," *Biostatistics*, vol. 25, no. 1, pp. 147–163, Jan. 2024.
<https://doi.org/10.1093/biostatistics/kxaf007>

APPENDIX

Dataset Name: WHO_COVID_Excess_Deaths_Estimates_By_Countries.xlsx

Dataset Link: <https://www.who.int/data/sets/global-excess-deaths-associated-with-covid-19-modelled-estimates>

Project GitHub Link:

https://github.com/PrathamAgrawal51/Pratham_Agrawal_22070521078_ML_CA1

Deployed Dashboard Link:

<https://pratham-covidexcessdeathsdashboard.streamlit.app/>

Name: Pratham Agrawal

PRN:22070521078

Sem: 7th

Sec: C